Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

Pablo Paramio Pérez
CEMFI Master in Economics and Finance

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Abstract

Glossary of acronyms

Acronym	Meaning
CCGT	Combined-cycle gas turbine: a flexible, dispatchable thermal generator.
DA	Day-ahead wholesale electricity market, cleared once per day.
IDA	Intraday auctions: sequence of (European) auctions held after the day-ahead clearing.
MTU	Market time unit: size of the time interval at which bids are submitted and prices clear.
REE	Red Eléctrica de España, the Spanish transmission-system operator.
OMIE	Operador del Mercado Ibérico de Energía, the market operator for the Spanish and Portuguese wholesale electricity markets.

1 Introduction

1.1 Motivation and research question

1.2 Related literature

1.3 Mechanism summary and contribution

TWO DISTINCT PURPOSES OF MTU15:

- OPERATIONAL / SYSTEM-OPERATOR CHANNEL. LETTING THE SYSTEM FOLLOW WITHIN-HOUR SOLAR VARIABILITY (CLOUD COVER, RAMP). MIDDAY IS EXACTLY WHERE THIS MATTERS MOST.
- STRATEGIC-EXPLOITATION CHANNEL. LETTING PIVOTAL FIRMS RE-PRICE ACROSS QUARTERS WHEN NET-LOAD MOVES SYSTEMATICALLY WITHIN THE HOUR. THIS IS WHAT DOMINANT CCGT CAN MONETIZE. **THIS IS WHAT WE DO!!**

1.4 Roadmap

2 Theoretical model

2.1 Setup

2.2 Equilibrium under low granularity

2.3 Equilibrium under high granularity

2.4 Predictions: dominant vs fringe response in time-varying vs flat subperiods

TODO: COROLLARY ON RAMP-CONSTRAINED HOURS. WHEN THE SYSTEM OPERATOR IMPOSES BINDING RAMP RESTRICTIONS (QUANTITY CANNOT ADJUST MUCH ACROSS CONSECUTIVE QUARTERS), THE HIGH-GRANULARITY REGIME FORCES FIRMS TO AMPLIFY THEIR STRATEGIC RESPONSE ON THE PRICE MARGIN RATHER THAN THE QUANTITY MARGIN. THE MTU15 REFORM THEREFORE AMPLIFIES MARKET POWER IN PRECISELY THOSE HOURS WHERE PHYSICAL ADJUSTMENT IS CONSTRAINED.

3 Institutional setting, data, and identification

3.1 Spanish wholesale electricity market: a sequential-market overview

STYLISED MARKET DIAGRAM TBD HERE AS TIKZ: DA - IDA AUCTIONS - CONTINUOUS - BALANCING

3.2 The MTU15 reform sequence

3.3 Data sources

OMIE

- Day-ahead data:

- Intraday-auction data:
- Continuous intraday market data:

ESIOS
ENTSO-E
INCLUDE SAMPLE SIZE AND FAMILIES.

3.4 Critical-vs-flat hours: empirical calibration

Hours are classified by the **within-hour variance of residual demand**,

$$\sigma^2_{\text{within}}(h) \;=\; \mathbb{E}_d\left[\frac{1}{4}\sum_{q=1}^{4}\big(D_{h,q,d}-\bar{D}_{h,d}\big)^2\right], \qquad D_{h,q,d}=\text{demand}_{h,q,d}-\text{wind}_{h,q,d}-\text{solar}_{h,q,d}. \tag{1}$$

Critical hours: high σ^2_{within}, 05:00–09:00 and 16:00–23:00 (eleven clock-hours). **Flat hours**: low σ^2_{within}, 01:00–04:00 (three clock-hours). Borderline hours are dropped. Full 24-hour ranking in Appendix B.6.

OJO, IF WE INTRODUCE THE 11:00-15:00 PERIOD IN THE CONTROL, WE ARE MIXING THE OPERATIONAL AND STRATEGIC CHANNELS OF THE REFORM, BECAUSE THE OPERATIONAL EFFECT IS CONCENTRATED IN THOSE HOURS!!!

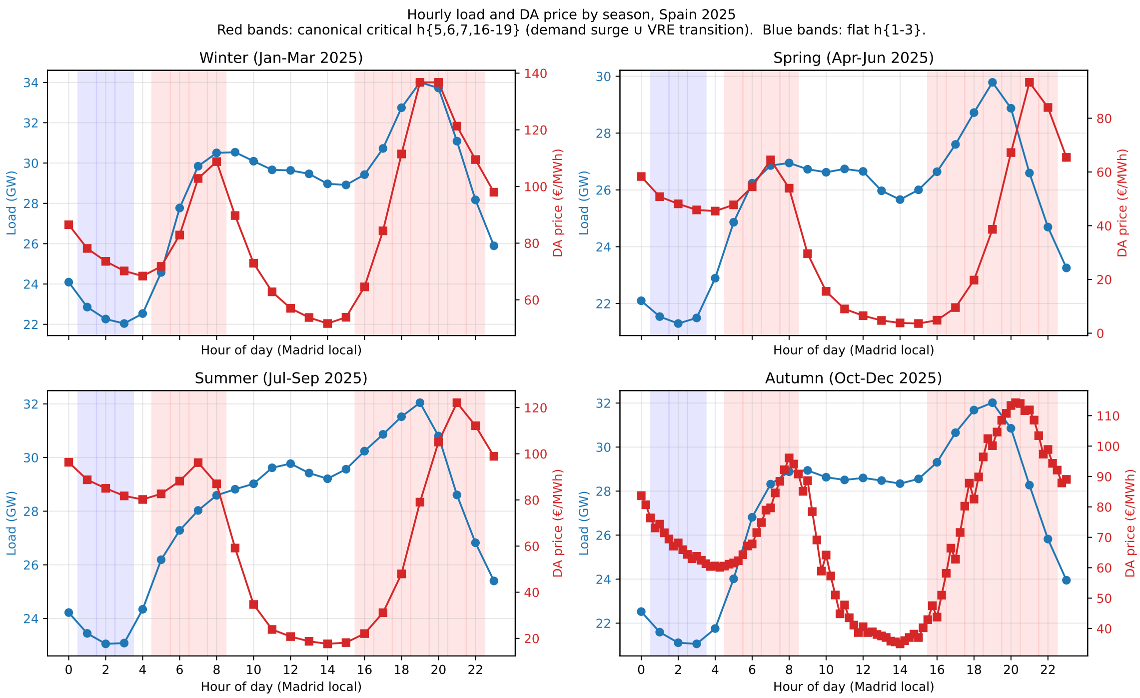


Figure 1: Hourly load and DA price by season, Spain 2025. Blue: demand; red: DA clearing price. Red bands: critical hours (05:00–09:00, 16:00–23:00); blue bands: flat hours (01:00–04:00).

Reading (**supportive**). Critical hours combine the morning ramp and evening peak; flat hours fall in the overnight valley. The pattern is stable across seasons.

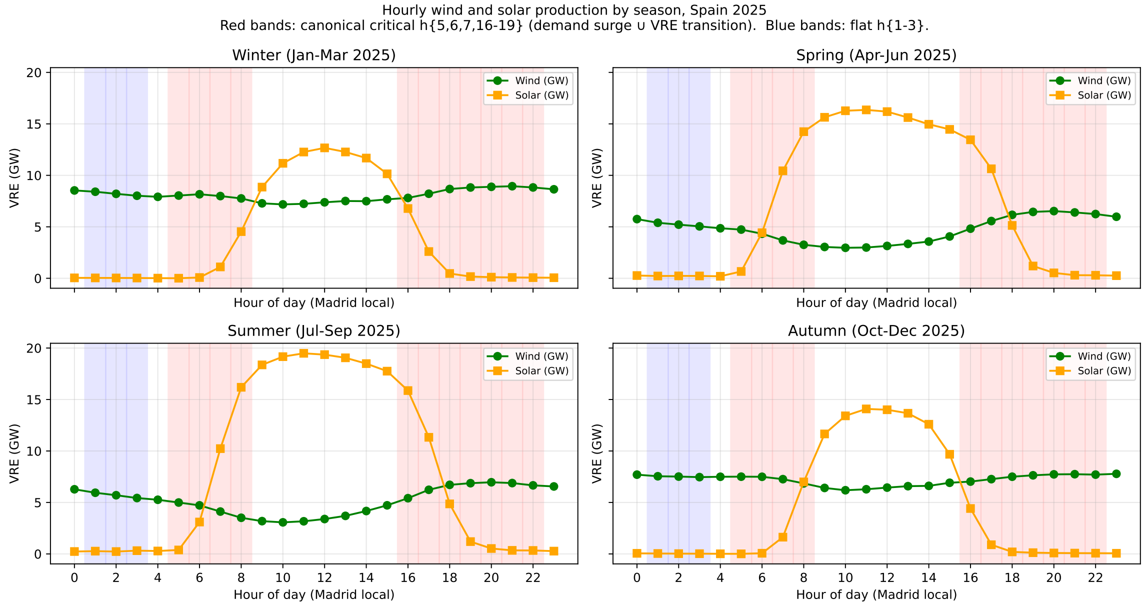


Figure 2: Hourly wind and solar production by season, Spain 2025. Red bands: critical hours; blue bands: flat hours.

Reading (**supportive**). Solar production concentrates around midday and fades sharply at the critical-hour boundaries; wind is broadly flat. Critical hours systematically face low renewable supply, reinforcing their role as the time-varying window.

3.5 Firms in the analysis sample

MENTION THAT WE USE LISTA_UNIDADES AND LISTA_AGENTES FROM OMIE, AS WELL AS THE SPECIFIC FAMILIES FORM OMIE, WHICH HAS FIRM AND UNIT INDICATORS. ALSO IMPORTANT TO MENTION THE CNMC CLASSIFICATION OF DOMINANT (SOURCE)

The empirical analysis follows nine parent firms in two blocks. The first block contains the *dominant operators in Spanish generation* (the CNMC *operador-dominante-en-generación* tier): Iberdrola, Endesa, Naturgy and EDP-Spain. EDP-Portugal joins this block as the dominant Iberian generator on the Portuguese side of MIBEL. The second block contains four other large vertically-integrated operators – Repsol, Engie España, TotalEnergies, and Moeve – which retain substantial portfolios but are non-dominant in the CNMC sense. Whether the firms in each block actually have a strategic margin in critical hours is an empirical question that we answer in Section 4.1, examining pivotality on the day-ahead bid data.

Table 1: Firms in the analysis sample. Capacity: max DA-offered power across 2025, share-weighted. Market share: PDBF programmed sell, 2025. Channel breakdown in Appendix B.2.

Parent firm	Main generation technologies	Capacity (GW)	Market share (%)
Dominant operators in Spanish generation (CNMC tier)			
Iberdrola	Solar, Hydro, CCGT	31.03	24.94
Endesa	Hydro, CCGT, Nuclear	12.46	16.39
Naturgy	CCGT, Hydro, Solar	12.60	12.34
EDP-Spain	Wind, Hydro, Solar	4.73	4.14
EDP-Portugal	Hydro, CCGT	7.42	15.17
Other large operators (non-dominant)			
Repsol	Solar, Hydro	0.61	1.43
Engie España	Wind, Solar, CCGT	4.05	1.73
TotalEnergies	Solar, CCGT, Wind	1.19	0.63
Moeve	Solar, Hydro, CCGT	1.28	0.01

Reading (**descriptive**). The four CNMC-dominant operators (Iberdrola, Endesa, Naturgy, EDP-Spain) plus EDP-Portugal together hold the bulk of Iberian generation capacity. The non-dominant block (Repsol, Engie, TotalEnergies, Moeve) is substantially smaller and serves as the placebo.

3.6 Identification strategy and treatment-effect set

WE SHOULD DEFINE HERE HOW WE DEFINE PIVOTALITY AND HOW WE CALCULATE IT, WITH DISCUSSION.

ALSO WE SHOULD DISCUSS THAT THE TREATMENT IN OUR IDENTIFICATION IS AT THE HOUR LEVEL (CRITICAL VS FLAT), NOT AT THE FIRM LEVEL, AND THAT FIRMS ENTER THE ANALYSIS AS SUBJECTS: PIVOTAL FIRMS ARE THOSE EXPECTED TO REACT TO THE TREATMENT BECAUSE THEY HAVE A STRATEGIC MARGIN IN CRITICAL HOURS; NON-PIVOTAL FIRMS SERVE AS PLACEBO BECAUSE THEY BID AT MARGINAL COST REGARDLESS OF HOUR.

HOWEVER, WE SHOULD THINK IF THIS IS THE BEST WAY TO PRESENT IT, MAYBE IT IS BETTER TO TALK ABOUT "EXPOSURE" TO THE TREATMENT ACROSS TIME OR SOMETHING LIKE THAT!! TO THINK ABOUT IT

3.7 Parallel-trends discussion

IMPORTANT THING HERE IS TO DISCUSS IT FOR EACH OF THE OUTCOMES AND TO TALK ABOUT CONDITIONAL PARALLEL TREND WITH RENEWABLE CAPACITY CONTROLS!

4 Bidding-behavior evidence of the mechanism

4.1 Pivotality: which firms can exploit the granularity reform

OJO, HYDRO DOES NOT EXPLOIT GRANULARITY BECAUSE THE DISPATCH DECISION IS DOMINATED BY RESERVOIR WATER-VALUE OPTIMIZATION (A MULTI-DAY PROBLEM), NOT WITHIN-HOUR STRATEGY. SO THE GRANULARITY REFORM'S "STRATE-

GIC EXPLOITATION CHANNEL" HAS NO PURCHASE HERE EVEN THOUGH THE CAPACITY FOR MARKET POWER EXISTS!!!!

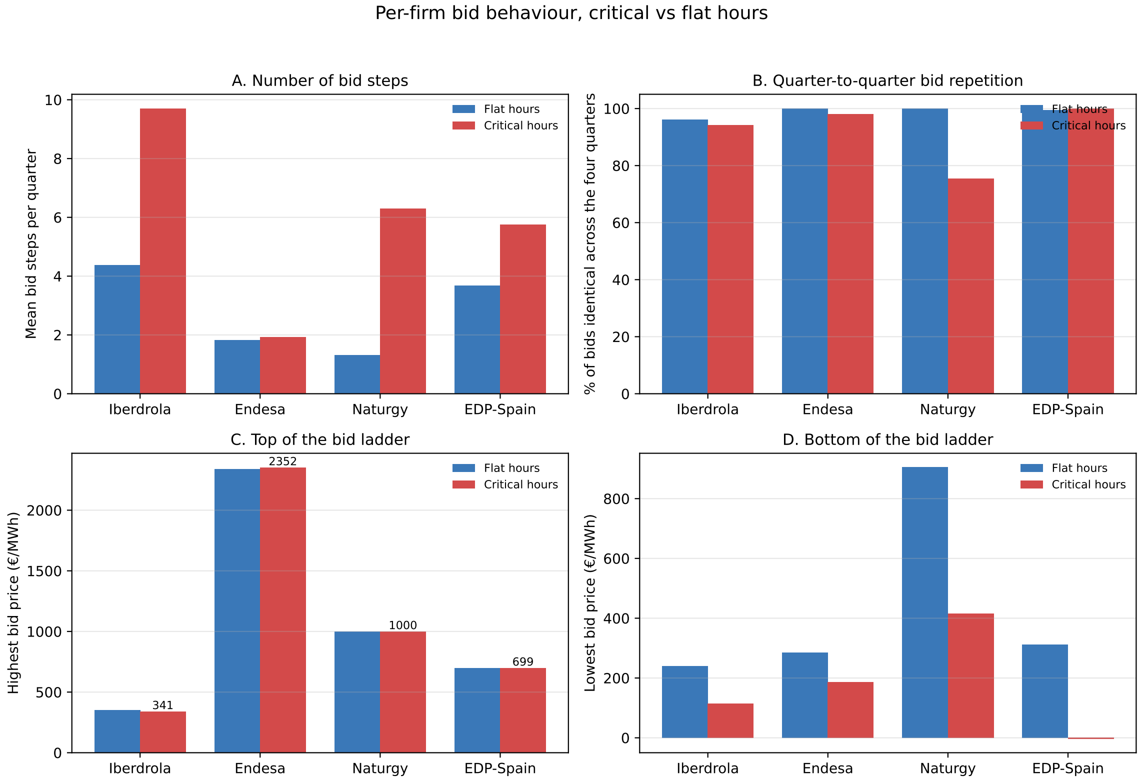
ONLY CCGT EXPLOITS GRANULARITY!! AND IN PARTICULAR NATURGY!

Table 2: Pivotality by parent firm and hour-class, CCGT, October–December 2025. Pivotality = share of (date, unit, period) cells with a bid tranche within 1 EUR/MWh of DA clearing.

Firm (parent)	flat (01:00–04:00)	critical (05:00–09:00, 16:00–23:00)	Δ crit−flat
Tightly pivotal in critical hours (within 1 EUR/MWh of clearing)			
Naturgy	0.006	0.044	**0.038**
EDP-Spain	0.026	0.041	**0.015**
EDP-Portugal	0.005	0.028	0.023
Iberdrola	0.003	0.012	0.009
Endesa	0.001	0.004	0.003
Essentially never tightly pivotal (negative control)			
Engie España	0.000	0.000	0.000
Moeve (formerly Cepsa)	0.000	0.000	0.000
Repsol	0.000	0.000	0.000
TotalEnergies	0.000	0.000	0.000

Reading (**supportive**). Naturgy is the most pivotal firm in critical hours (4.4%), followed by EDP-Spain and EDP-Portugal. Repsol, Engie, TotalEnergies, and Moeve are essentially never tightly pivotal and serve as the negative control.

4.2 Per-firm bid-shape and granularity exploitation



A bid step is one price-quantity pair in a firm's supply offer for a given period; a firm with more steps offers a finer-grained supply curve. 'Identical across quarters' = the firm submitted the exact same step (price and quantity) in all four quarters of the same clock-hour. Sample: combined-cycle gas (CCGT) units operated by the four largest Iberian groups, day-ahead market, October--December 2025 (post-reform window).

Figure 3: Per-firm DA bid behaviour, critical vs flat hours. Top-4 CCGT operators, October–December 2025. Panel A: number of price-quantity steps per 15-minute period. Panel B: share of bids identical across all four quarters of the same hour. Panels C–D: highest and lowest priced step.

Reading (**supportive**). Two distinct strategic patterns appear. Iberdrola enriches the bid ladder in critical hours (more steps) but keeps the four quarters of each hour nearly identical. Naturgy keeps the ladder shallow and varies bids strongly across quarters.

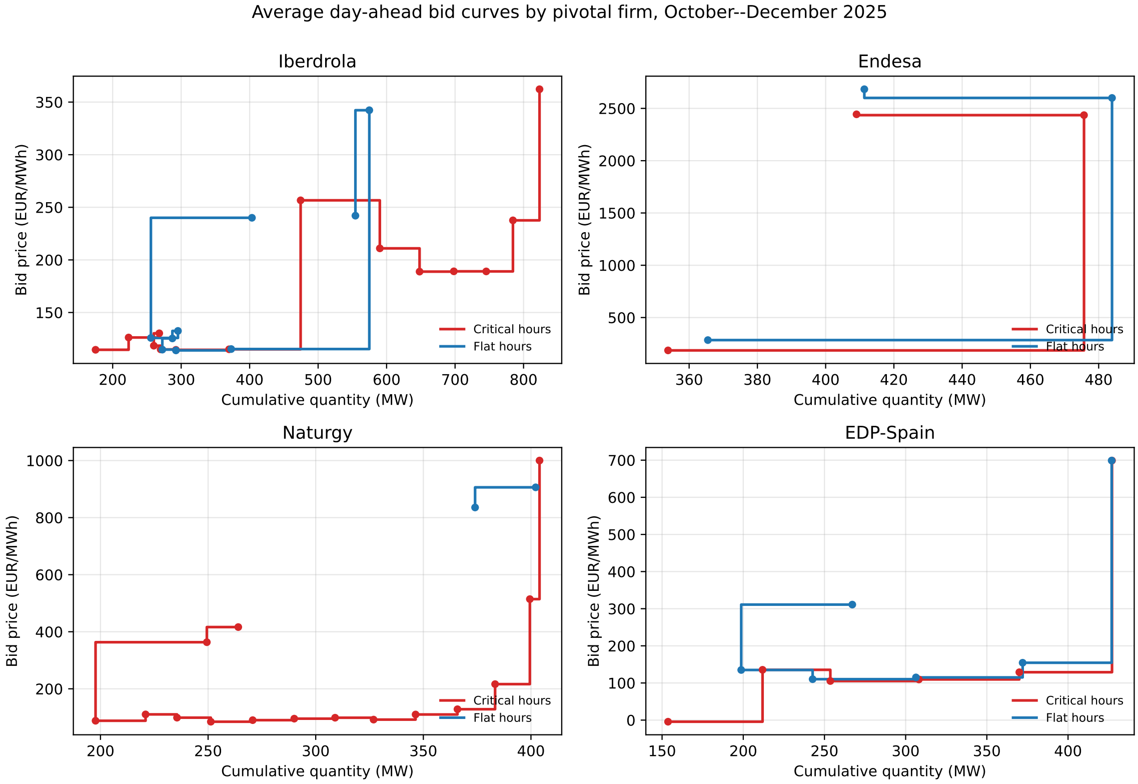


Figure 4: Average DA bid curves by pivotal firm, October–December 2025. Step functions are average price-quantity ladders across (date, unit, period) cells, indexed by tranche rank. Red: critical hours; blue: flat hours.

Reading (**supportive**). The bid curves in critical hours sit visibly lower and span a wider quantity range than in flat hours for Iberdrola, Naturgy and EDP-Spain — pivotal firms put more capacity in play at lower prices when the granularity matters. Endesa is the exception: an essentially flat ladder at very high prices that does not shift between hour classes.

Table 3: Per-firm DA bid-shape detail, October–December 2025. Mean number of tranches, median tranche price, total quantity offered, and the share of clock-hours in which all four 15-min quarters carry an identical bid (mechanical-repeat rate).

Firm	Hour-class	Mean tranches	Median price (EUR/MWh)	Total qty (MW)	Mechanical repeat rate	N
Iberdrola	Critical	9.70	134.8	612.4	0.990	31,324
	Flat	4.38	245.7	614.4	0.969	8,556
Endesa	Critical	1.92	1090.0	460.7	0.982	20,780
	Flat	1.82	1179.4	460.6	1.000	5,652
Naturgy	Critical	6.30	538.1	410.4	0.811	59,720
	Flat	1.32	945.5	410.4	1.000	16,260
EDP-Spain	Critical	5.76	121.1	426.8	1.000	8,052
	Flat	3.68	375.9	424.9	1.000	2,196
EDP-Portugal	Critical	6.60	153.0	407.0	0.075	18,556
	Flat	5.52	233.0	407.0	0.270	5,064

Reading (**mixed**). The ladder enrichment in critical hours is clear in Iberdrola (9.7 vs 4.4 tranches) and Naturgy (6.3 vs 1.3); Endesa stays flat (1.9 vs 1.8). Mechanical repetition across the four quarters drops sharply in critical hours only for Naturgy (0.81 vs 1.00) and EDP-Portugal (0.07 vs 0.27): they are the firms genuinely exploiting MTU15 within-hour granularity. Iberdrola, Endesa, and EDP-Spain enrich the ladder but copy it across quarters.

5 Main empirical results

5.1 Strategic withholding and intraday upward repositioning

MECHANISM: UNDER THE HIGHER-GRANULARITY REGIME, A PIVOTAL DOMINANT FIRM IN A CRITICAL HOUR CAN WITHHOLD CAPACITY FROM THE DAY-AHEAD CLEARING AND REPOSITION THAT CAPACITY UPWARD IN THE INTRADAY AUCTIONS AT HIGHER PER-QUARTER PRICES. UPWARD SELL ADJUSTMENT SUMMED ACROSS THE THREE INTRADAY AUCTIONS, $q_{2,fdh}$, IN MWH PER UNIT-CLOCK-HOUR:

$$q_{2,fdh} \;=\; \alpha + \beta_1\,\mathrm{crit}_h + \beta_2\,\mathrm{post}_d + \beta_3\,(\mathrm{crit}_h \times \mathrm{post}_d) + \gamma_f + \delta_{\mathrm{DOW}(d)} + \varepsilon_{fdh}, \tag{2}$$

with firm fixed effects γ_f, day-of-week fixed effects $\delta_{\mathrm{DOW}(d)}$, and cluster-robust standard errors by date. The sample restricts to the same-calendar-month windows October–December 2024 (pre) and October–December 2025 (post), so the March 2025 intraday reform is held constant. Critical hours are 05:00–09:00 and 16:00–23:00; flat hours are 01:00–04:00. Pivotal firms are those with non-trivial price-setting capacity in critical hours (Iberdrola, Endesa, Naturgy, EDP-Spain, EDP-Portugal); non-pivotal firms (Repsol, Engie España, TotalEnergies, Moeve) serve as a negative control.

Table 4: B1 – Within-day DiD on q_2 (intraday upward adjustment). Estimates of Equation 2. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	All firms (1)	Pivotal firms (2)	Non-pivotal firms (3)
Critical (β_1)	1.280***	-4.454***	4.993***
	(0.328)	(0.893)	(0.320)
Post (β_2)	-2.562***	-8.294***	1.140**
	(0.486)	(1.146)	(0.415)
Critical × Post (β_3)	-1.397**	3.866***	-4.741***
	(0.446)	(1.147)	(0.423)
Observations	293,838	122,495	171,343
Clusters (days)	184	184	184
R^2	0.0132	0.0056	0.0113
Adj. R^2	0.0131	0.0055	0.0113

Reading (**supportive**). $\beta_3 = +3.87$ MWh per unit-clock-hour for pivotal firms and -4.74 for the non-pivotal placebo, both significant at the 0.1% level. The sign reverses across firm classes, which the placebo prediction requires.

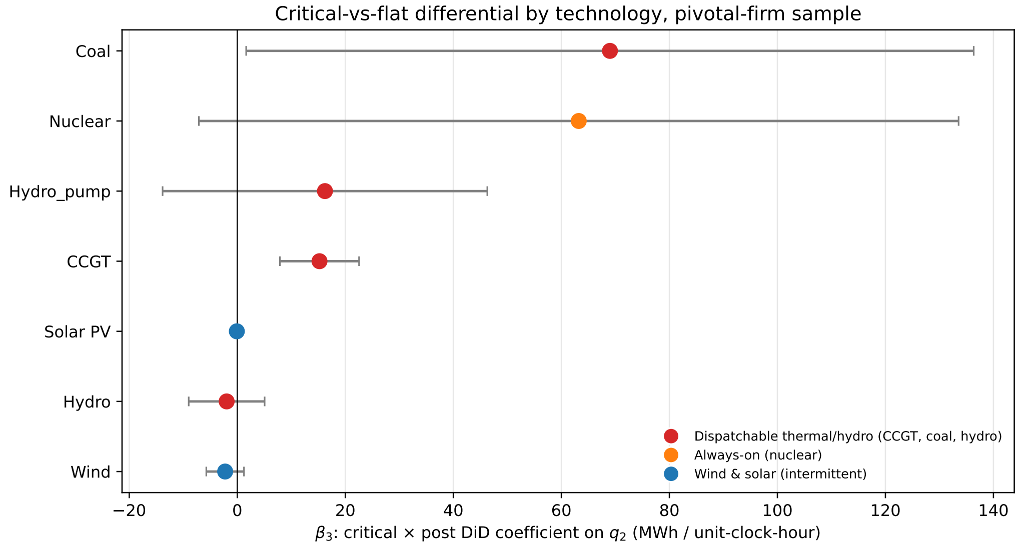


Figure 5: B1 – β_3 by technology, pivotal sample. Equation 2 restricted by `tech_group`; 95% cluster-robust CIs.

Reading (**mixed**). Positive β_3 concentrates in the strategically dispatchable techs (CCGT +15.2, Coal +69, Hydro_pump +16.2); Wind and Solar PV both show the predicted null. Hydro is mildly negative against a positive prediction, consistent with reservoir water-value optimisation rather than within-hour strategy. Nuclear is large with very wide error bars and few units — neither sharply contradicts the mechanism nor cleanly supports it.

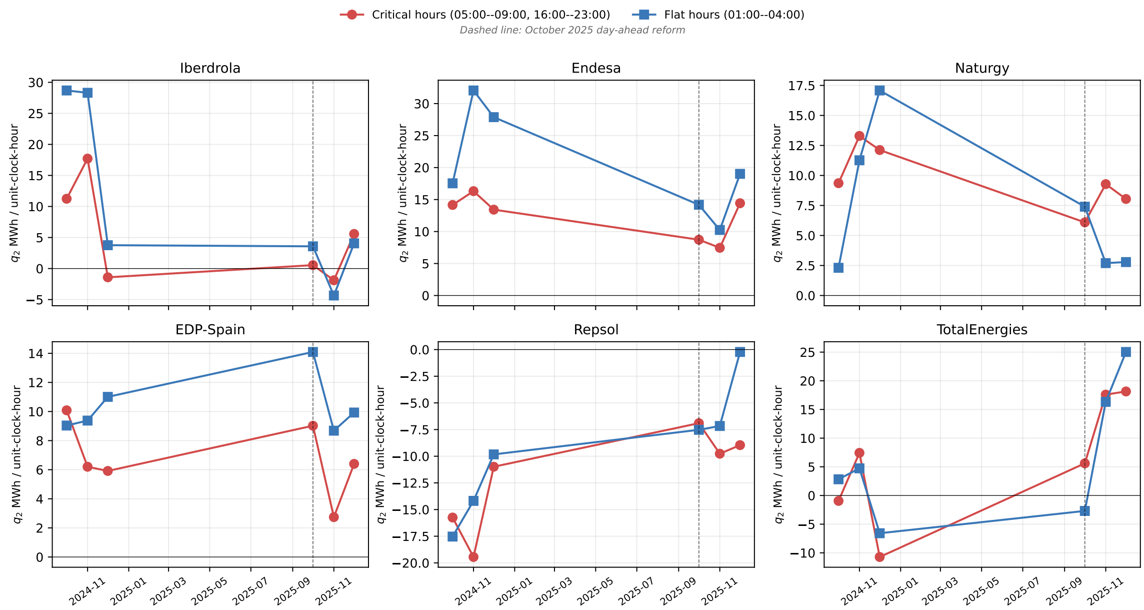


Figure 6: q_2 monthly mean by firm and hour-class. Red: critical (05:00–09:00, 16:00–23:00); blue: flat (01:00–04:00). Dashed line: October 2025 reform.

Reading (**supportive**). The pivotal-firm panels show the critical line tracking above the flat line after October 2025, while the non-pivotal panels show the two lines collapsing together. The pre-reform trends are visually parallel in both blocks.

The full table of β_3 stratified by technology (with the non-pivotal placebo column) is reported in Appendix A.4. The per-firm decomposition of β_3 jointly on q_2 and on day-ahead cleared volume (the withholding outcome) is in Appendix A.5. A supporting analysis on the day-ahead

→ intraday price wedge is in Appendix A.6; the wedge is a noisy outcome under the post-blackout *operación reforzada* regulatory overlay and is not informative on its own.

5.2 Parallel trends discussion

$$\mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \,|\, \text{crit}] \;=\; \mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \,|\, \text{flat}]\,. \tag{3}$$

6 Conclusion

A Robustness checks

A.1 Sensitivity to critical-hours definition

A.2 Sensitivity to firm-partition (pivotality vs administrative)

A.3 Same-cal-month vs full-window comparisons

Table 5: B5 – β_3 across sensitivities, pivotal firms on q_2. Each row re-estimates Equation 2 under one variation: critical-hours definition (B5.1), firm partition (B5.2), window (B5.3/B5.5), sample exclusions (B5.4), DST transition days (B5.6/B5.7).

Sensitivity	β_3	SE	p	G
B5.1 — Critical-hours definition				
supply_ramp h{7,8,16,17,18}	4.28***	(1.26)	0.0007	184
price_peak h{18-22}	5.68***	(1.49)	0.0001	184
demand_peak h{16-20}	5.13***	(1.36)	0.0002	184
joint h{7-8, 16-22}	4.77***	(1.31)	0.0003	184
B5.2 — Firm partition				
Pivotality-based firm set	3.87***	(1.15)	0.0008	184
Administrative IB/GE/GN/HC	3.87***	(1.15)	0.0008	184
B5.3 / B5.5 — Window				
Full window 2024-2025	3.42***	(0.80)	0.0000	730
Full window minus Apr-Sep 2025	4.00***	(0.85)	0.0000	575
B5.4 — Sample exclusions				
Drop EDP-PT	3.87***	(1.15)	0.0008	184
Drop ABO2G	3.87***	(1.15)	0.0008	184
B5.6 — DST transition days (CET ↔ CEST)				
Same-cal-month, drop DST transition days	4.00***	(1.15)	0.0005	182
Full window, drop DST transition days	3.47***	(0.80)	0.0000	726
B5.7 — DST regime separation (clock-hour semantics differ across CEST/CET)				
Same-cal-month, CEST days only (Oct 1–25)	−2.45	(1.93)	0.2028	50
Same-cal-month, CET days only (Oct 27–Dec 31)	6.30***	(1.34)	0.0000	132

Reading (**supportive**). β_3 stays in the +3.4 to +6.3 MWh range across critical-hours definitions, firm partitions, window choices, and DST handling, with $p < 0.001$ throughout. The only sub-window where the sign flips is the 50-day CEST-only segment, which is small and confounded by daylight saving.

A.4 β_3 stratified by technology

Table 6: B1 – β_3 by technology, pivotal sample. Equation 2 restricted by `tech_group`.

	CCGT	Hydro	Hydro pump	Coal	Nuclear	Wind	Solar PV
Critical × Post (β_3)	15.23***	-1.963	16.24	69.00*	63.22	-2.244	0
	(3.738)	(3.586)	(15.34)	(34.36)	(35.88)	(1.779)	(.)
Obs.	25,232	16,997	3,125	413	2,440	31,344	7,276
Clusters	184	184	183	58	118	184	184

Reading (**mixed**). Same picture as Figure 5 in numeric form: dispatchable carry the result, Wind and Solar PV are null, Nuclear is large but extremely noisy. Hydro pushes against the directional prediction, in line with reservoir water-value optimisation rather than within-hour strategy.

A.5 Per-firm decomposition: q_2 and day-ahead cleared MWh

Table 7: Per-firm β_3 on q_2 and DA cleared MWh.

	q_2 outcome (B1)		DA cleared (B3)	
Firm	β_3	SE	β_3	SE
Iberdrola	10.40**	(3.25)	−62.37***	(14.02)
Endesa	7.30*	(3.06)	−2.22	(6.57)
Naturgy	2.58	(1.79)	−34.69***	(7.69)
EDP-Spain	−2.29	(1.53)	−0.31	(3.33)
EDP-Portugal	—	—	−24.18*	(11.43)

Reading (**mixed**). Iberdrola and Naturgy show the joint pattern predicted by the mechanism (negative DA, positive IDA). Endesa adds in IDA without DA withholding, and EDP-Spain is weak on both sides — the mechanism is not uniform across pivotal firms. EDP-Portugal lacks `pibci` data so only the DA side is identified for that firm.

A.6 DA → IDA price wedge (supporting outcome)

Table 8: B2 – Within-day DiD on the DA-IDA price wedge. Outcome in EUR/MWh.

Specification	β_3	SE	p	G
Same-cal-month (Oct-Dec 2024 vs 2025)	4.62**	(1.43)	0.0012	184
Full window, no controls	0.79	(1.18)	0.5043	730
Full window, + reforzada × crit	−10.06***	(1.39)	0.0000	730
Full window, + reforzada + MTU15-IDA × crit	−10.06***	(1.39)	0.0000	730

Reading (**problematic**). Under same-calendar-month restriction the within-day wedge expands modestly post-reform, but once the reforzada × critical interaction is included in the full-window spec the reform's own coefficient turns large and negative. The wedge is not informative without controlling for the regulatory overlay, so we cannot use it as a clean supporting outcome.

A.7 Operational vs strategic decomposition (PIBCA vs PHF)

A.8 Bid-shape robustness in the competitive-zone restriction

A.9 Time placebos

A.10 Triple-difference: pivotality as moderator

$$
\begin{aligned}
q_{2,fudh} = \alpha + \beta_1\,\mathrm{crit}_h + \beta_2\,\mathrm{post}_d + \beta_3\,\mathrm{piv}_f \\
+ \beta_{12}\,\mathrm{crit}_h\,\mathrm{post}_d + \beta_{13}\,\mathrm{crit}_h\,\mathrm{piv}_f + \beta_{23}\,\mathrm{post}_d\,\mathrm{piv}_f \\
+ \beta_{123}\,\mathrm{crit}_h\,\mathrm{post}_d\,\mathrm{piv}_f + \gamma_f + \delta_{\mathrm{DOW}(d)} + \boldsymbol{\theta}'\,X_{dh} + \varepsilon_{fudh}, \quad (4)
\end{aligned}
$$

where $X_{dh} = (\mathrm{wind}_{dh}, \mathrm{solar}_{dh}, \mathrm{demand}_{dh})$ in GW.

Table 9: Triple-difference on q_2. Equation 4; pivotal and non-pivotal firms pooled. SE clustered by date.

	(1) DDD baseline	(2) + X (additive)	(3) + X×{crit, post, piv}	(4) + clock-hour FE
β_{123} (crit×post×piv)	8.605	8.623	8.612	8.619
	(1.362)	(1.364)	(1.361)	(1.364)
Implied β_3, pivotal firms	3.862	3.874	4.006	3.875
	(1.146)	(1.147)	(1.154)	(1.148)
Implied β_3, non-pivotal firms	-4.743	-4.748	-4.605	-4.743
	(0.423)	(0.422)	(0.488)	(0.422)
Observations	293,838	293,838	293,838	293,838

Reading (**supportive**). $\beta_{123} \approx +8.6$ MWh (SE $\approx$ 1.4) across all four specifications, with $p < 0.001$. The covariates and clock-hour fixed effects move the coefficient by less than 0.02 MWh, so the result does not rely on the day-level controls.

Table 10: Time-placebo β_3 on q_2. Equation 2 re-estimated over windows not crossing the October 2025 reform. P1/P2 are within-regime placebos; P3 crosses the June 2024 intraday reform.

	P1: within-2024			P2: within-2025 pre-reform			P3: shifted back 1 year		
	All	Pivotal	Non-piv.	All	Pivotal	Non-piv.	All	Pivotal	Non-piv.
Critical × Post (β_3)	0.942	-0.0305	1.910***	-0.0637	1.200	-1.133***	-0.916	-2.785*	0.199
	(0.490)	(1.235)	(0.440)	(0.407)	(0.894)	(0.280)	(0.536)	(1.341)	(0.439)
Obs.	253,079	98,092	154,987	327,300	146,896	180,404	250,320	99,636	150,684
Clusters	184	184	184	182	182	182	184	184	184

Reading (**supportive**). P1 (within-2024) and P2 (within-2025 pre-MTU15-DA) give β_3 statistically indistinguishable from zero in the pivotal sample. P3, which crosses the June 2024 intraday reform, picks up a negative coefficient consistent with that earlier structural shift, not the October 2025 reform.

B Data appendix

B.1 Sequential-market technical reference (PDBC, PDBF, PDVD, PIBCA, PHF)

B.2 Firms in the analysis sample (detailed)

I THINK WE CAN TAKE THE CAPAPCITY DATA! BUT MAYBE IT IS ONLY FOR EXPORTS AND IMPORTS!.

Column definitions. *DA auction share*: PDBF offer-type 1, anonymous DA clearing. *DA bilateral share*: PDBF offer-type 4, pre-negotiated firm-to-firm transfers. *IDA cumulative share*: PIBCA accumulated (last session per unit-period). *Net position*: net seller if 2025 DA sales $>$ 60% of turnover, net buyer if $<$ 40%, mixed otherwise. The net-position label is descriptive; vertical integration does not enter the identification.

Table 11: Firms in the analysis sample (detailed). Channel-segregated 2025 shares. Channel denominators are channel-specific (they do not sum across rows).

Parent firm	Gen. units	Capacity (GW)	DA auction share (%)	DA bilateral share (%)	IDA cumul. share (%)	Net position
Dominant operators in Spanish generation (CNMC tier)						
Iberdrola	70	31.03	20.01	27.18	20.37	Mixed
Endesa	38	12.46	36.88	7.11	12.64	Net buyer
Naturgy	39	12.60	3.86	16.18	7.67	Net buyer
EDP-Spain	21	4.73	0.77	5.67	2.60	Net buyer
EDP-Portugal	16	7.42	20.62	12.70	5.17	—
Other large operators (non-dominant)						
Repsol	13	0.61	1.79	1.26	0.62	Net buyer
Engie España	59	4.05	0.03	2.49	1.07	Net seller
TotalEnergies	15	1.19	1.19	0.38	0.59	Net buyer
Moeve	112	1.28	0.03	—	0.66	Net buyer

Reading (**descriptive**). Iberdrola is roughly balanced across channels. Endesa is auction-heavy. Naturgy is bilateral-heavy. Vertical-integration status is descriptive only and does not enter the identification.

B.3 File-format specifications and parser conventions

B.4 Power-vs-energy unit discipline

B.5 Coverage and missingness

B.6 Hour-of-day classification metrics

Table 12: Hour-of-day classification metrics, Spain 2025. Mean load, solar, net-load (= load − wind − solar), within-hour SD of net-load, and signed within-hour load change $\Delta\text{load} = \text{load}_{q_4} - \text{load}_{q_1}$, by clock-hour.

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	Δ demand (MW)	Class
00:00–01:00	23.2	0.2	7.1	16.0	405	−864	*dropped*
01:00–02:00	22.4	0.1	6.8	15.4	353	−481	*flat*
02:00–03:00	21.9	0.1	6.6	15.1	269	−193	*flat*
03:00–04:00	21.9	0.1	6.5	15.3	324	194	*flat*
04:00–05:00	22.9	0.1	6.4	16.4	649	1179	*dropped*
05:00–06:00	24.9	0.3	6.3	18.3	766	1795	*critical*
06:00–07:00	27.0	1.9	6.2	18.9	1009	1420	*critical*
07:00–08:00	28.3	5.9	5.8	16.7	1259	602	*critical*
08:00–09:00	28.7	10.5	5.3	12.9	1116	218	*critical*
09:00–10:00	28.7	13.6	5.0	10.1	783	−103	*dropped*
10:00–11:00	28.6	15.0	4.8	8.7	576	−78	*dropped*
11:00–12:00	28.6	15.6	4.9	8.1	471	116	*dropped*
12:00–13:00	28.7	15.6	5.1	8.0	448	−60	*dropped*
13:00–14:00	28.3	15.2	5.3	7.9	453	−290	*dropped*
14:00–15:00	28.0	14.4	5.5	8.1	545	−57	*dropped*
15:00–16:00	28.3	13.0	5.8	9.4	747	344	*dropped*
16:00–17:00	28.9	10.1	6.3	12.5	1241	634	*critical*
17:00–18:00	30.0	6.4	6.8	16.8	1362	902	*critical*
18:00–19:00	31.2	2.7	7.3	21.2	1185	847	*critical*
19:00–20:00	32.0	0.7	7.4	23.9	557	163	*critical*
20:00–21:00	31.1	0.3	7.5	23.2	614	−1487	*critical*
21:00–22:00	28.6	0.2	7.5	20.9	801	−1926	*critical*
22:00–23:00	26.4	0.2	7.4	18.8	694	−1441	*critical*
23:00–00:00	24.6	0.2	7.2	17.2	534	−1093	*dropped*

Reading (**supportive**). Critical hours (05:00–09:00, 16:00–23:00) have within-hour SD of net-load several times higher than flat hours (01:00–04:00). The ranking pins the classification directly to the structural variability of residual demand.

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